$\Pi = \top, \sigma = \emptyset$

unsigned int x = 1;
unsigned int y = b;

$\Pi = \top, \sigma = \{\phi_1 = 1, \phi_2 = \alpha_2\}$
$x \mapsto \phi_1, y \mapsto \phi_2$

while (y < 4)

$\Pi = (\phi_2 \geq 4), x \mapsto \phi_1, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2\}$

assert y >= x;}

$\Pi = (\phi_2 \geq 4) \wedge (\phi_2 \geq \phi_1) \mid x \mapsto \phi_1, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2\}$

—

$\Pi = (\phi_2 \geq 4) \wedge (\phi_2 < \phi_1) \mid x \mapsto \phi_1, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2\}$

—

$\Pi = (\phi_2 < 4), x \mapsto \phi_1, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2\}$

x = x + a

$\Pi = (\phi_2 < 4), x \mapsto \phi_3, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1\}$

if (x >= 1)

$\Pi = (\phi_2 < 4) \wedge (\phi_3 < 1), x \mapsto \phi_3, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1\}$

break;}}, assert (y >= x);}

$\Pi = (\phi_2 < 4) \wedge (\phi_3 < 1) \wedge (\phi_2 \geq \phi_3) \mid x \mapsto \phi_3, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1\}$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 < 1) \wedge (\phi_2 < \phi_3) \mid x \mapsto \phi_3, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1\}$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1), x \mapsto \phi_3, y \mapsto \phi_4$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\}$

while (y < 4)

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1), x \mapsto \phi_3, y \mapsto \phi_4$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\} \mid x \mapsto \phi_3, y \mapsto \phi_4$

x = x + a

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\} \mid x \mapsto \phi_3, y \mapsto \phi_4$

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1\} \mid x \mapsto \phi_5, y \mapsto \phi_4$

if (x >= 1)

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1\} \mid x \mapsto \phi_5, y \mapsto \phi_4$

break;}}, assert (y >= x);}

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1\} \mid \wedge(\phi_4 \geq \phi_5)\ x \mapsto \phi_5, y \mapsto \phi_4$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1\} \mid \wedge(\phi_4 < \phi_5)\ x \mapsto \phi_5, y \mapsto \phi_4$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1\} \mid x \mapsto \phi_5, y \mapsto \phi_4$

y = x + b - 1

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1,$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1, \phi_6 = \phi_4 + \alpha_2 - 1\} \mid x \mapsto \phi_5, y \mapsto \phi_6$

break;}}, assert (y >= x);}

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1) \wedge (\phi_6 \geq \phi_5)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1,$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1, \phi_6 = \phi_4 + \alpha_2 - 1\} \mid x \mapsto \phi_5, y \mapsto \phi_6$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 < 4) \wedge (\phi_5 < 1) \wedge (\phi_6 \geq \phi_5)$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1,$
$\phi_4 = \phi_3 + \alpha_2 - 1, \phi_5 = \phi_3 + \alpha_1, \phi_6 = \phi_4 + \alpha_2 - 1\} \mid x \mapsto \phi_5, y \mapsto \phi_6$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1), x \mapsto \phi_3, y \mapsto \phi_2$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1\}$

y = x + b - 1

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 \geq 4), x \mapsto \phi_3, y \mapsto \phi_4$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\}$

assert y >= x;}

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 \geq 4) \wedge (\phi_4 \geq \phi_3) \mid x \mapsto \phi_3, y \mapsto \phi_4$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\}$

—

$\Pi = (\phi_2 < 4) \wedge (\phi_3 \geq 1) \wedge (\phi_4 \geq 4) \wedge (\phi_4 < \phi_3) \mid x \mapsto \phi_3, y \mapsto \phi_4$
$\sigma = \{\phi_1 = 1, \phi_2 = \alpha_2, \phi_3 = \phi_1 + \alpha_1$
$\phi_4 = \phi_3 + \alpha_2 - 1\}$

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